

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

COMPUTER SCIENCE

Examination, May 2022

Nature-Inspired Computation

Module Leader: Dr Zeliang Wang

Duration: TWO HOURS

Answer ALL questions.

The marks for this module are calculated from 60% of the percentage mark for this paper plus 40% of the percentage mark for associated coursework.

No Word Count Specified

This is an CLOSED BOOK examination.

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Question 1

- (a) Briefly describe the key algorithmic elements of an evolutionary algorithm (EA), and include feasible parameter settings for each of the parameters associated with the algorithm.

(10 marks)

- (b) There are two types of representations used in evolutionary algorithms, i.e., *direct* and *indirect* representations. Explain the difference between them and provide an example for each type of representations.

(6 marks)

- (c) Explain what are meant by the terms *exploration* and *exploitation* in the context of evolutionary algorithms?

(4 marks)

- (d) Describe the main differences between *genetic algorithm* and *genetic programming*.

(6 marks)

- (e) What is swarm intelligence? Give three examples of swarm behaviour that demonstrates the inspiration to computing.

(5 marks)

- (f) When training a neural network, the weights can be initialised as zero values. Do you think this is a good initialisation strategy? Describe your reasoning.

(5 marks)

- (g) Conway's Game of Life is probably the most famous cellular automaton. What are the major rules used to represent the state of a cell? Explain under what circumstances a particular rule is triggered.

(4 marks)

(Total 40 marks)

Question 2

- (a) Considering a genetic algorithm in which individuals are represented using a 6-bit string of the form $b_1b_2b_3b_4b_5b_6$. An example of an individual is 001011 for which $b_1 = 0$, $b_2 = 0$, $b_3 = 1$, $b_4 = 0$, $b_5 = 1$, $b_6 = 1$. The *fitness function* for these individuals is defined by

$$f(b_1b_2b_3b_4b_5b_6) = b_1 + b_2 + b_3 + b_4 + b_5 + b_6 + \text{AND}(b_1, b_2, b_3, b_4, b_5, b_6),$$

where $\text{AND}(b_1, b_2, b_3, b_4, b_5, b_6) = 1$ if $b_1 = b_2 = b_3 = b_4 = b_5 = b_6 = 1$, and $\text{AND}(b_1, b_2, b_3, b_4, b_5, b_6) = 0$ otherwise.

- (i) Write down the fitness values and the probabilities of selecting each of the individuals according to Roulette Wheel selection method for the following individuals and show each step of your work.

- 001011
- 101011
- 000000
- 100100
- 111111

(10 marks)

- (ii) Assume that a single point crossover will be applied to the individuals, and this point has been chosen as the point between the 3rd and the 4th bits (i.e. between b_3 and b_4). Show the two offspring that will result from crossing over the following two individuals.

- First parent: 001011
- Second parent: 101001

(4 marks)

- (iii) Name the most appropriate mutation operator for this representation and briefly explain how this mutation method is applied after selection and crossover to form the next generation of a population.

(4 marks)

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(b) Suppose you have been asked to develop a workload scheduling solution for an airline company using genetic algorithms. There are 5 planes and 8 cabin crews in the company. Each plane can have only one cabin crew on a single day, and each crew cannot work for more than two days in a row. The company uses all planes every day. Your task is to work out the best combination of crews on any particular day using a genetic algorithm.

(i) Describe how you would represent an individual using a chromosome in this algorithm.

(2 marks)

(ii) Describe how you would formulate a fitness function for this problem.

(4 marks)

(iii) How many solutions are there for this problem? Do you think it is necessary to use genetic algorithms for solving it? What if the company operated more planes and employed more crews?

(6 marks)

(Total 30 marks)

Question 3

- (a) Suppose we would like to train a perceptron for a binary classification problem using the following training data:

Input (x_1)	Input (x_2)	Target (y)
0	0	0
1	1	1
1	-1	0

The initial weight vector and bias are given by $W_0 = [-1, 0]$ and $b_0 = 0.5$. The activation is a step function φ which takes the value 1 when the input is greater than or equal to 0, and 0 otherwise.

- (i) Sketch the initial decision boundary based on W_0 and label the input datapoints. Which of the three input patterns are correctly classified using the initial weights and bias?

(5 marks)

- (ii) Train the perceptron using the perceptron learning rule, i.e., present each input vector **once** in the order shown and find the new decision boundary. Show the steps of your work.

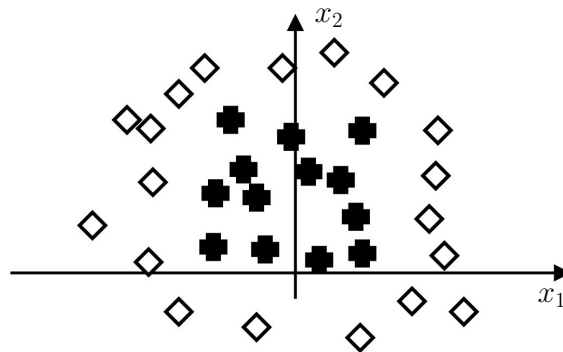
(8 marks)

- (iii) Sketch the new decision boundary after the previous training step, and again indicate which of the three input patterns are now correctly classified.

(6 marks)

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- (b) Given a set of training data $X = [x_1, x_2]$ as depicted in the following diagram, we want to build a neural network that is able to classify these two dimensional data into two classes (i.e., crosses and diamonds).



- (i) Draw a diagram of a neural network that can solve this classification problem.

(5 marks)

- (ii) Justify your choice of the architecture and the number of nodes in each layer.

(6 marks)

(Total 30 marks)